
Professional Summary:

- Senior Data Engineer with 7 years of enterprise experience delivering large-scale AI/ML, Generative AI, and MLOps solutions across Banking, Healthcare, Oil & Gas, and E-commerce domains.
- Designed end-to-end AI pipelines across AWS, GCP, and Azure, integrating SageMaker, Vertex AI, Bedrock, Azure OpenAI, and Dataiku DSS, enabling cross-cloud training, deployment, and governance.
- Built retrieval augmented generation (RAG) pipelines with LangChain, Pinecone, FAISS, Weaviate, and LlamaIndex, powering semantic search engines and contextual reasoning systems used in banking and healthcare.
- Implemented Agentic AI frameworks including LangGraph, AutoGen, CrewAI, and Agentic Swarm to orchestrate autonomous multi-agent systems, enabling adaptive reasoning, fraud investigation, and compliance workflows.
- Applied MCP (Model Context Protocol) to securely connect LLMs with APIs, enterprise systems, and compliance datasets, improving AI governance and reliability.
- Fine-tuned LLMs with RLHF and LoRA techniques, aligning GPT, LLaMA, Claude, and FLAN-T5 with domain-specific financial and healthcare requirements.
- Delivered fraud detection and risk scoring systems using XGBoost, LightGBM, Random Forest, and LSTMs, achieving 22%+ precision improvement and 40% reduction in false positives.
- Engineered synthetic dataset generation workflows using GANs and LLM augmentation, enhancing rare-event recall by 17% in healthcare and BFSI models.
- Implemented AI Ops monitoring pipelines with automated anomaly detection, drift alerts, and self-healing triggers, reducing model downtime by 30%.
- Hands-on expertise in developing Snowflake Snowpark stored procedures using Java for advanced data transformations and rule-based data processing.
- Experienced in building modular, reusable data components and validation frameworks in Java, integrating with Snowflake and AWS Glue.
- Strong background in optimizing data pipelines, ensuring real-time processing through Java-based APIs, Kafka producers/consumers, and Airflow orchestration.
- Applied OTEL observability for real-time distributed tracing, logging, and metrics across Kubernetes-based ML microservices.
- Built and maintained Fivetran-based pipelines for automated, real-time and batch ingestion across multiple domains.
- Experienced in text parsing, regex-based linguistic processing, and ontology-driven NLP systems.
- Skilled in rule-based NLP pipelines alongside statistical and vector-based methods.
- Strong expertise in AWS Glue + Iceberg pipelines for scalable, compliant enterprise data integration.
- Hands-on experience with metadata management, lineage tracking, and data cataloging tools.
- Provided Tier 2/3 support for Fivetran sync issues, Glue job failures, and Iceberg inconsistencies, ensuring high reliability of data flows.
- Architected real-time ML inference services with FastAPI, Celery, Lambda, and ECS/EKS, reducing scoring latency by 30–40% in production workloads.
- Designed CI/CD pipelines with Terraform, Jenkins, GitHub Actions, Docker, and Helm, ensuring seamless deployments with rollback safety and version control.
- Drafted detailed, executable natural language ML implementation plans for complex projects, ensuring reproducibility and traceability.

- Hands-on experience with benchmarking ML models and pipelines for performance and training speed improvements.
- Designed AI/ML solutions with identity-aware data access controls, ensuring that only authorized roles could query sensitive datasets.
- Proficient in Docker-based ML development environments, ensuring consistent and portable experimentation.
- Focused on reproducibility in ML research, documenting experiments and validating results against original plans.
- Strong experience with independent, compliance-driven ML development workflows.
- Migrated Hadoop workloads to Iceberg, Trino, and Flink, improving query performance, governance, and cost efficiency by 30%+.
- Created BI dashboards in Tableau, Power BI, and Looker embedding ML predictions, KPIs, and explainability metrics, enabling faster executive decision-making.
- Developed cross-cloud MLOps governance frameworks integrating Dataiku DSS, SageMaker, Redshift ML, and Vertex AI for end-to-end lifecycle management.
- Implemented knowledge graph reasoning pipelines to support compliance reviews, delivering faster insights and contextual fraud analysis.
- Applied Bayesian optimization and hyperparameter tuning (Optuna, Hyperopt) to improve model robustness and reduce training overhead.
- Designed, deployed, and maintained production-grade Elasticsearch clusters with high availability, scalability, and fault-tolerant configurations.
- Implemented indexing and shard/replica strategies to optimize query speed and data retrieval.
- Developed Logstash pipelines and configured Beats agents for real-time ingestion of structured/unstructured financial datasets.
- Hands-on expertise in developing Snowflake Snowpark stored procedures using Java for advanced data transformations and rule-based data processing.
- Experienced in building modular, reusable data components and validation frameworks in Java, integrating with Snowflake and AWS Glue.
- Strong background in optimizing data pipelines, ensuring real-time processing through Java-based APIs, Kafka producers/consumers, and Airflow orchestration.
- Built Kibana dashboards and visualizations for monitoring, anomaly detection, and compliance insights.
- Troubleshoot node failures, GC pauses, and slow queries, ensuring consistent cluster performance.
- Applied RBAC and TLS encryption to secure Elasticsearch clusters.
- Experience applying ELK Stack to Capital Markets, supporting trading data analysis, risk monitoring, and financial transaction pipelines.
- Designed domain-specific NLP solutions (SpaCy, Hugging Face, GPT APIs) for entity recognition, summarization, PII detection, and medical text classification.
- Built multi-region, fault-tolerant architectures on AWS/GCP for sub-second inference and 24x7 fraud detection reliability.
- Delivered cloud cost savings of 35%, 40% faster ML deployments, and 20% fewer ETL failures across enterprise-scale systems.
- Created synthetic patient and fraud datasets to accelerate ML adoption while ensuring HIPAA and GDPR compliance.
- Designed explainability-first AI workflows with SHAP and LIME, cutting compliance approval turnaround by 25%.
- Built interactive fraud alerting chatbots with Amazon Lex and Vertex AI conversational APIs, reducing customer verification time by 35%.
- Integrated autonomous AI agents with Elasticsearch, SQL/NoSQL DBs, Jira, Confluence, Git to unify structured and unstructured data pipelines.

- Developed backend automation services in Python, supporting data ingestion, monitoring, and enterprise integration.
- Applied Infrastructure as Code (Terraform & Ansible) for automated provisioning, scaling, and compliance.
- Implemented CI/CD pipelines with Git, GitHub Actions, and Jenkins, standardizing version control and automated testing workflows.
- Collaborated in Agile teams, ensuring iterative delivery of AI/MLOps features in enterprise environments.
- Partnered with cross-functional teams to bridge engineering, analytics, compliance, and product, ensuring smooth AI adoption in regulated environments.
- Mentored engineers and analysts in LangGraph, RAG, MLOps best practices, and GenAI architectures, building team-wide AI capabilities.

Technologies:

Category	Technologies
Languages & Scripting	Python, SQL, PL/SQL, R, Java, Scala, PySpark, Bash, Shell, TypeScript, JavaScript (ES6+), HTML5, CSS3, JSON, YAML, XML
Machine Learning & AI	SageMaker, Bedrock, Vertex AI, Azure ML, TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, Redshift ML, Hugging Face, LangChain, Claude, Falcon, Mistral, LLaMA-2/3, LoRA, RLHF, LlamaIndex, SHAP, LIME, Time Series (ARIMA, Prophet, LSTM), Recommenders, NLP (SpaCy, NLTK, Gensim), RAG Pipelines
Agentic AI & GenAI	LangGraph, AutoGen, CrewAI, Agentic Swarm, MCP Protocol, Prompt Engineering (Few-shot, Chaining, Templates), Synthetic Data (GANs, LLM Augmentation)
Data Engineering	Apache Spark, Flink, Kafka, Hive, Hadoop, Talend, Airflow, DBT, Glue, EMR, EKS, Kinesis, Athena, Delta Lake, Parquet, ORC, Avro, Snowflake, BigQuery, Elasticsearch, Dataflow, Pub/Sub, Cloud Composer, Vector DBs (FAISS, Pinecone, Weaviate), Fivetran, Apache Iceberg
Cloud & DevOps	AWS (S3, EC2, Lambda, SageMaker, Redshift, Glue, Bedrock, CodePipeline, CloudWatch), GCP (BigQuery, Vertex AI, Dataflow, Pub/Sub, Looker, GKE), Azure (ADF, Synapse, Databricks, Purview, OpenAI), Terraform, Jenkins, GitHub Actions, Docker, Kubernetes, Helm, ArgoCD, Redis
Databases & Warehousing	Oracle 11g/12c/19c, Oracle EBS CRM, PostgreSQL, SQL Server, MySQL, MongoDB, DynamoDB, Cassandra, Neo4j, TimescaleDB, Snowflake, Redshift, BigQuery, SAP HANA
Visualization & Reporting	Tableau, Power BI, Looker, Superset, Data Studio, Streamlit, Dash, ReactJS, Flask, FastAPI
Security & Compliance	IAM, KMS, RBAC, VPC SC, Data Masking, Encryption, GDPR, HIPAA, CCPA, SOC2, OTEL Observability, CloudTrail, Audit Logging
Modeling & Optimization	Star/Snowflake Schema, Data Vault 2.0, OLAP, Bayesian Optimization, Hyperparameter Tuning, PSI Drift Detection, Feature Engineering & Selection, Lineage & Governance

Professional Experience:

Senior Data Engineer with MLOps

Oct 2023 – Present

Bank of America, Dallas, TX

Responsibilities:

- Designed and deployed asynchronous ML inference pipelines using FastAPI, Celery, AWS Lambda, and ECS, reducing fraud scoring latency by 35% and enabling real-time decisioning.
- Built hybrid ML workflows across AWS SageMaker and GCP Vertex AI, improving training-to-deployment cycles by 25% and standardizing model governance across clouds.
- Fine-tuned fraud detection LLMs (Bedrock + Hugging Face) using LoRA and RLHF, improving fraud detection precision by 22% and reducing false positives in high-risk segments.
- Developed knowledge graph-augmented reasoning pipelines for compliance workflows, enhancing contextual investigation and improving review turnaround by 30%.
- Implemented RAG retrieval systems with LangChain + FAISS, giving analysts real-time access to fraud case histories and reducing investigation time by 40%.
- Designed LangGraph agentic workflows with branching logic, memory persistence, and contextual reasoning, automating multi-agent collaboration in fraud investigations.
- Designed Elasticsearch cluster configurations with optimized shard/replica strategies to improve query and indexing performance.
- Developed regex-based parsers for financial and healthcare text normalization. Applied ontology alignment techniques with knowledge graph reasoning for fraud/compliance text analytics.
- Implemented rule-based tokenization and hybrid NLP (regex + ML) for medical terminology.
- Developed Logstash pipelines to ingest fraud and compliance datasets from heterogeneous sources into Elasticsearch.
- Developed Java-based Snowpark stored procedures to perform complex data transformations and quality checks directly within Snowflake, reducing ETL latency by 25%.
- Built custom Java libraries for metadata validation, error handling, and schema enforcement integrated with AWS Glue and Snowflake.
- Implemented Java microservices for event-based ingestion pipelines using Kafka and Snowpipe, enabling real-time data synchronization across platforms.
- Designed and optimized Snowflake Snowpark UDFs (User Defined Functions) using Java for reusable computation logic in analytics workflows.
- Integrated Snowflake with AWS S3 and Lambda through Java SDKs to automate ingestion and transformation processes.
- Configured Beats agents for streaming logs and system telemetry into Elasticsearch clusters.
- Built Kibana dashboards for real-time fraud case analysis and compliance monitoring.
- Implemented cluster troubleshooting and tuning practices to resolve query slowness, node failures, and indexing bottlenecks.
- Integrated MCP (Model Context Protocol) for secure communication between LLMs, APIs, and enterprise databases, ensuring compliance in model interactions.
- Built Pinecone-backed vector search APIs for semantic queries in fraud detection pipelines, enabling sub-second retrieval of fraud intelligence.
- Automated PSI-based drift detection triggers in Dataiku, SageMaker, and Vertex AI, initiating retraining pipelines and reducing model degradation by 25%.
- Orchestrated synthetic dataset generation workflows using GANs and LLM augmentation to improve rare-event fraud recall by 17%.

- Migrated Hadoop pipelines to Iceberg + Trino, improving data query speed and governance while reducing infrastructure costs by 30%.
- Integrated Elasticsearch with AI pipelines to index structured + unstructured fraud case histories for rapid retrieval.
- Built Fivetran connectors for automated ingestion of fraud and compliance datasets, integrating with Glue jobs and Iceberg tables.
- Configured and monitored AWS Glue workflows and Iceberg table formats for scalable governance and analytics.
- Delivered Tier 2/3 pipeline support, resolving Fivetran sync failures, Glue errors, and Iceberg inconsistencies.
- Built backend Python automation services to streamline fraud investigation workflows and model monitoring.
- Enhanced enterprise AI adoption by integrating pipelines with Jira and Confluence, improving auditability and collaboration.
- Drafted natural language technical plans for ML workflows, ensuring engineers could execute reproducible implementations.
- Validated Python implementations against documented plans, flagging discrepancies and improving model traceability.
- Conducted benchmarking of fraud detection ML models, comparing training speed and performance across clouds.
- Worked extensively in Docker-based ML environments, standardizing reproducible experiments for fraud analytics.
- Applied Ansible alongside Terraform to automate infrastructure provisioning for fraud detection microservices.
- Integrated Core.ID identity management with AI fraud detection pipelines to enforce secure role-based model access.
- Configured SSO and MFA policies via Core.ID for analysts and developers accessing AI dashboards and fraud case datasets.
- Partnered with compliance teams to ensure identity governance policies aligned with regulatory requirements (CCPA, GDPR, SOC2).
- Implemented Nessie/Polaris catalog management for versioned ML lineage and reproducibility across fraud pipelines.
- Deployed CI/CD pipelines with Helm, Jenkins, and GitHub Actions, reducing release times by 30% and ensuring rollback safety.
- Applied OTEL observability in Python/Kubernetes ML microservices, improving distributed tracing, latency monitoring, and anomaly detection across fraud services.
- Engineered multi-region AWS architectures for fault-tolerant fraud detection with sub-second inference, ensuring 24/7 high availability.
- Delivered fraud adoption dashboards in Tableau and Excel for stakeholders, enabling visibility into model usage, anomalies, and error trends.
- Embedded explainability (SHAP + LIME) into dashboards and reports, accelerating compliance approval cycles by 25%.
- Built generative AI pipelines (GPT-4, FLAN-T5, LangChain) to summarize fraud cases and investment reports, increasing analyst productivity by 28%.
- Applied RLHF + reward modeling to optimize LLM outputs for compliance-driven fraud narratives, ensuring accuracy in high-stakes reporting.
- Developed interactive fraud alerting chatbots using Amazon Lex and Vertex AI conversational APIs, reducing customer verification time by 35%.
- Integrated CrewAI + AutoGen multi-agent workflows, automating fraud case investigation through distributed AI agent collaboration.

- Mentored engineers on LangGraph best practices, including graph design, branching, memory optimization, and deployment at scale.
- Partnered with compliance and business teams to track adoption metrics and improve fraud model accuracy across multiple portfolios.
- Conducted A/B testing and Bayesian inference for fraud model feature validation, boosting predictive accuracy and stability.
- Built MLflow experiment tracking and artifact versioning, enabling smooth lifecycle management for fraud models.
- Delivered synthetic anomaly detection datasets to regulators as evidence of robust fraud risk management processes.

Data Engineer II

Sempra, Los Angeles, CA

Dec 2021 – Sep 2023

Responsibilities:

- Designed and deployed ML pipelines in Talend integrated with AWS SageMaker and GCP Vertex AI, cutting model training time by 33% and streamlining MLOps.
- Orchestrated automated retraining workflows with Talend, Lambda, and Dataflow, improving forecasting model accuracy by 16%.
- Built Dataiku DSS workflows for energy demand prediction, integrated with S3 + BigQuery, and automated PSI-based drift detection, boosting forecast accuracy by 14%.
- Developed containerized inference microservices (FastAPI, Celery, EKS, GKE) for real-time energy forecasting, reducing deployment cycles by 28%.
- Designed executable ML workflow plans for energy forecasting use cases, bridging documentation with code execution.
- Engineered Java data orchestration components to coordinate Snowflake Tasks, Streams, and Snowpipe operations for continuous ingestion.
- Created modular Java data connectors for Talend pipelines, improving throughput between AWS and Snowflake by 20%.
- Designed and deployed Airflow DAGs triggering Java-based Snowflake Snowpark transformations for scheduled data refreshes.
- Applied Core.ID federation with Azure AD to secure access to energy forecasting models and datasets.
- Enforced identity-based restrictions in ML pipelines, ensuring operational data access adhered to governance frameworks.
- Built Logstash-based ingestion pipelines for high-volume sensor and energy demand datasets.
- Created Kibana visualizations to monitor demand anomalies and operational metrics.
- Tuned Elasticsearch indexing and queries for performance at scale, reducing latency in real-time forecasting analytics.
- Designed real-time data replication pipelines with Fivetran, integrating S3, Glue, and Iceberg for energy forecasting.
- Documented data flows and architectural decisions, enabling reproducibility and compliance audits.
- Benchmarked forecasting pipelines to compare model speed and reproducibility, improving reliability of demand predictions.
- Implemented containerized solutions in Docker for portable ML deployments and testing.
- Prototyped agentic swarm forecasting systems in LangGraph and AutoGen, coordinating multiple AI agents to adapt predictions to fluctuating demand scenarios.
- Migrated Hadoop pipelines to Iceberg + Flink, cutting compute costs by 30% and significantly improving scalability and query performance.
- Designed and deployed Trino-based query engines for high-performance analytics on Iceberg tables.

- Leveraged AutoML (SageMaker Autopilot + Vertex AI) for feature engineering, reducing experimentation time by 35%.
- Integrated SHAP + LIME explainability into BigQuery ML forecasting models, improving stakeholder approval cycles by 25%.
- Created executive dashboards in Tableau and Excel to track adoption, forecasting accuracy, and energy demand patterns.
- Deployed Elasticsearch connectors for high-volume energy data search and retrieval.
- Standardized CI/CD and Git-based version control for ML pipeline deployments.
- Partnered with cross-functional teams in an Agile environment, ensuring incremental delivery of energy analytics solutions.
- Configured OTEL + Prometheus/Grafana to monitor latency, throughput, and error rates across containerized energy models.
- Integrated SAP HANA and Oracle EBS CRM datasets into Redshift and BigQuery, reducing reporting cycle times by 40%.
- Implemented CI/CD pipelines with GitHub Actions, Terraform, and Helm, lowering rollback times by 25%.
- Applied IAM + KMS-based controls for sensitive energy and customer data, improving compliance audit scores by 20%.

Data Engineer

Sep 2020 – Nov 2021

MedTourEasy, New Delhi, India

Responsibilities:

- Built HIPAA-compliant ETL pipelines using Talend + Glue to ingest EHR, clinical, and claims data from APIs, Salesforce Health Cloud, and flat files into Redshift and BigQuery.
- Developed healthcare risk scoring models in Dataiku DSS, automating KPI dashboards and integrating with Redshift/Glue pipelines for patient retention tracking.
- Applied NLP models (Hugging Face Transformers + GPT APIs) in Dataiku to summarize patient engagement notes, improving medical staff productivity.
- Integrated Core.ID with healthcare data pipelines, securing patient EHR access in compliance with HIPAA guidelines.
- Built identity-aware AI workflows to ensure PII/PHI datasets could only be accessed by authorized medical practitioners.
- Developed Java APIs to automate patient data ingestion into Snowflake, integrating JDBC connections with Redshift and BigQuery sources.
- Applied object-oriented design principles in Java for data lineage tracking and auditing frameworks across healthcare data flows.
- Implemented AI Ops-driven monitoring with OTEL observability for healthcare ETL and inference pipelines, ensuring SLA compliance.
- Designed Excel and Tableau dashboards to track adoption and accuracy of patient risk scoring models.
- Engineered claims anomaly detection workflows integrating payer + provider datasets, reducing fraud leakage.
- Created step-by-step ML implementation plans for healthcare models, enabling compliance-ready reproducibility.
- Benchmarked anomaly detection models for accuracy vs. speed tradeoffs, ensuring optimized healthcare fraud detection.
- Fine-tuned transformer models with RLHF on HIPAA-compliant datasets for clinical summarization tasks.
- Designed Logstash parsing workflows to structure clinical and claims data for Elasticsearch indexing.

- Developed Kibana dashboards for healthcare teams to track patient risk scores and fraud detection metrics.
- Onboarded new healthcare data sources using Fivetran connectors, streamlining ingestion into Glue + Redshift pipelines.
- Applied metadata cataloging and lineage tracking to ensure HIPAA-compliant healthcare integrations.
- Built RAG pipelines with FAISS embeddings, providing clinicians with contextual patient histories at query time.
- Connected healthcare ML pipelines with Jira and Confluence to automate reporting, ticketing, and compliance tracking.
- Designed domain-specific tokenization strategies for medical terminology, improving NLP accuracy on clinical datasets.
- Packaged Python APIs into Helm-deployed Kubernetes services, standardizing release processes across environments.
- Applied data masking, encryption, and hashing using AWS Lambda, KMS, and Secrets Manager, ensuring PII/PHI privacy.
- Developed chatbots and dashboards with Flask on EC2, integrating Redshift and BigQuery backend datasets for real-time patient insights.
- Implemented event-driven ingestion (S3 triggers, SNS, Step Functions) for telemetry and lab data, enabling near real-time processing.
- Designed fraud detection ML models for claims data using structured + unstructured healthcare datasets.

Data Analyst

May 2019 – Aug 2020

Futures First, Gurugram, Haryana, India

Responsibilities:

- Built HIPAA-compliant ETL pipelines using Talend + Glue to ingest EHR, clinical, and claims data from APIs, Salesforce Health Cloud, and flat files into Redshift and BigQuery.
- Developed healthcare risk scoring models in Dataiku DSS, automating KPI dashboards and integrating with Redshift/Glue pipelines for patient retention tracking.
- Applied NLP models (Hugging Face Transformers + GPT APIs) in Dataiku to summarize patient engagement notes, improving medical staff productivity.
- Configured Core.ID-based authentication and role management for financial risk dashboards, ensuring secure trading data access.
- Applied identity-aware logging and audit trails to support compliance audits for financial data analytics.
- Implemented AI Ops-driven monitoring with OTEL observability for healthcare ETL and inference pipelines, ensuring SLA compliance.
- Built Kibana visualizations for financial trading datasets, improving transparency in anomaly detection and risk analytics.
- Developed Java APIs to automate patient data ingestion into Snowflake, integrating JDBC connections with Redshift and BigQuery sources.
- Applied object-oriented design principles in Java for data lineage tracking and auditing frameworks across healthcare data flows.
- Configured Beats agents to stream log data from trading systems into Elasticsearch clusters.
- Applied indexing and query optimization to accelerate retrieval of large-scale trading data.
- Built Fivetran-based ingestion workflows for replicating financial datasets into Snowflake and Redshift.
- Monitored pipeline health for latency, throughput, and data quality, proactively resolving sync failures.

- Designed Excel and Tableau dashboards to track adoption and accuracy of patient risk scoring models.
- Engineered claims anomaly detection workflows integrating payer + provider datasets, reducing fraud leakage.
- Drafted structured ML experiment plans and validated outputs against documented specifications for financial analytics.
- Applied benchmarking techniques to measure reproducibility and stability of trading data models.
- Fine-tuned transformer models with RLHF on HIPAA-compliant datasets for clinical summarization tasks.
- Built RAG pipelines with FAISS embeddings, providing clinicians with contextual patient histories at query time.
- Designed domain-specific tokenization strategies for medical terminology, improving NLP accuracy on clinical datasets.
- Packaged Python APIs into Helm-deployed Kubernetes services, standardizing release processes across environments.
- Applied data masking, encryption, and hashing using AWS Lambda, KMS, and Secrets Manager, ensuring PII/PHI privacy.
- Developed chatbots and dashboards with Flask on EC2, integrating Redshift and BigQuery backend datasets for real-time patient insights.
- Implemented event-driven ingestion (S3 triggers, SNS, Step Functions) for telemetry and lab data, enabling near real-time processing.
- Designed fraud detection ML models for claims data using structured + unstructured healthcare datasets.